INCH-POUND

MIL-D-82885
27 October 1993
SUPERSEDING
MIS-44427
11 September 1991

MILITARY SPECIFICATION

DIAZODINITROPHENOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the manufacture and acceptance of one type of diazodinitrophenol commonly referred to as dinol, diazol, or DDNP.

1.2 Classification. The diazodinitrophenol shall be particulate-rounded and irregular agglomerates with no needle crystals.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-46106 Adhesive-Sealants, Silicone, RTV, One-Component

MIL-A-48078 Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1376

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MIL-B-82647 Bag, Conductive Plastic, Heat-Sealable, Flexible

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection By Attributes

MIL-STD-1168 Ammunition Lot Numbering

DOD-STD-2101 Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation (October 1, 1989 revision)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2905 Standard Practice for Statements on Number of Specimens for Textiles

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Color (M101). The diazodinitrophenol shall be greenish yellow to brown in color when examined as specified in 4.7.2.

3.3 Form (M102). The diazodinitrophenol shall be particulate-rounded and irregular agglomerates with no needle crystals when examined as specified in 4.7.3.

3.4 Granulation (M103). The average particle size shall be 300 $\pm$ 50 microns with 3 percent maximum in the 27 to 105 micron range when tested as specified in 4.7.4.

3.5 Explosive efficiency (M104). The explosive efficiency test shall produce 8-millimeter (mm) minimum holes in a 0.125-inch lead plate when tested as specified in 4.7.5.

3.6 Bulk density (M105). The bulk density shall be 1.9 to 2.5 grams (g) when tested as specified in 4.7.6.

3.7 Workmanship (M106). The contractor shall implement procedures and controls to assure that the process and the product are not compromised by foreign material and contaminants or any other conditions which may degrade the composition. Determination of foreign materials shall be in accordance with 4.7.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. The first article sample shall be 1 pound of the diazodinitrophenol. The sample shall be obtained from the first production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained form the same sources of supply a will be used in regular production. The first article sample shall be subjected to all of the examinations and tests specified in 4.7.

4.3.1 First article sample rejection. If the first article sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. A lot shall consist of one or more batches of diazodinitrophenol produced by one manufacturer, in accordance with the same specification under one continuous set of operating conditions. Each lot shall consist of that quantity of diazodinitrophenol that has been subjected to the same unit chemical or physical process intended to make the final product homogenous. The lot shall comply with the provisions for submission of product samples as specified in MIL-STD-105. Lot number assignment shall be in accordance with MIL-STD-1168. When specified in the contract or purchase order (see 6.2), the provisions of MIL-A-48078 shall apply.

4.4.2 Examinations. The examinations and tests listed in 4.7 shall be performed on production batches and on inspection lots as defined in 4.4.1. Sample testing shall be in accordance with 4.6.2. The examinations and tests shall verify conformance with the requirements of sections 3 and 5 herein.

4.4.3 Sample size. Quality conformance inspection sample size shall be as specified in the contract or purchase order (see 6.2).

4.4.4 Sampling plans. Unless otherwise specified in the contract or purchase order (see 6.2), sampling plans and procedures for verifying major and minor characteristics shall be in accordance with Inspection Level II of MIL-STD-105.

4.5 Classification of characteristics. Characteristics verified by the inspection and tests specified herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristics. Requirements not annotated by the classification code symbol are classified as minor.

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4.6 Testing.

CAUTION

This specification covers sampling and testing of toxic and hazardous material. Accordingly, all applicable safety rules, regulations, and procedures must be followed in handling and processing the diazodinitrophenol. Disposal of all chemicals and hazardous materials must be in accordance with all local, state, and federal regulations.

4.6.1 Sampling. By means of a hand or a horn spoon, sufficient material to form a primary sample of approximately 100 g (dry weight) shall be removed from each barrel or drum in the lot. Primary samples shall be made up of equal portions taken from each bag in the barrel or drum. Each primary sample shall be blended on a smooth surface by mixing with a horn spatula. Water shall be added in case the sample appears to be dry. The sample shall be spread out and divided into squares approximately 1/2 inch on a side by means of a spatula. Small portions shall be taken from each square to make a retained portion of the primary sample of approximately 20 g (dry weight). The retained portion of each primary sample shall be placed in a tightly stoppered bottle so labeled that the barrel or drum from which the sample was taken can be easily identified. The remaining portions of the primary samples shall be thoroughly mixed, spread out, marked into squares and small portions taken from each until a composite sample of approximately 50 g (dry weight) is obtained. This composite sample shall be placed in a smooth-necked bottle with a tight-fitting stopper and labeled to show the name of the material, manufacturer, plant, contract or order number, lot number, and the number of pounds in the lot. All acceptance tests shall be made on the composite samples representative of the lot. The primary samples shall be held for possible future examination should the composite sample fail to meet the requirements of this specification.

4.6.2 Testing for lot acceptance. The sample obtained in 4.6.1 shall be tested in accordance with the applicable test methods of 4.7 for conformance to the requirements of 3.2 through 3.7.

4.6.3 Inspection equipment. Inspection equipment and its calibration, design, and approval shall be as specified in the contract or purchase order (see 6.2).

4.7 Test procedure. All tests shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Unless otherwise specified, all chemicals shall be reagent grade or ACS grade. It is recommended that all test procedures be documented by Standard Operating Procedures (SOPs) for organizational use. (See 6.3 for use of equivalent test methods.)

4.7.1 Preparation of dry sample. Since the sample as received will contain approximately 40 percent water by weight, the following procedure shall be followed to prepare dry material for those tests where it is needed. Place a piece of filter paper, cut to proper size, in a Gooch crucible held in an adapter on a suction flask. Transfer approximately 3 to 5 g of the sample to the crucible with a stream of water from a wash bottle. Air dry the sample by means of suction and then transfer the material to a paper tray. Spread the material to a thin layer, thereby breaking up lumps and then heat in a hot air oven at 60°C for a minimum of 16 hours. Hold the dried sample in a desiccator or in a rubber stoppered bottle until used.

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4.7.2 Color. The color shall meet the requirements of 3.2.

4.7.3 Form. Spread a thin layer of the sample over approximately a 0.5 square inch area of the central surface of a 1-inch by 3-inch glass microscope slide, in such a manner that the individual crystals are discernible. Form shall meet the requirements of 3.3 when viewed under the microscope.

4.7.4 Granulation. The average particle size shall be determined using the following procedure:

a. Six screens shall be assembled as shown on figure 1. The screen mesh sizes shall be 27, 105, 224, 317, 416, and 592 microns.

b. Mark screens and holding pans with screen mesh size and dry in oven for a minimum of 2 hours.

c. Cool and store screens and pans in a desiccator.

d. Weigh screens in matched pans to nearest 0.001 g and record tare weight.

e. Place screens and funnel with stopper on vacuum flask. Place O-ring on screen bottoms and wet the screen and O-ring assemblies with water. Stack the screens on top of screen funnel with the smallest screen on bottom progressing to the largest screen on the top (see figures 2 and 3).

f. Transfer approximately 1 g of diazodinitrophenol into the top screen.

g. Alternately wash the diazodinitrophenol through the screens with water from a wash bottle and apply vacuum.

h. When separation is complete in the top filter screen, remove excess water with vacuum. Remove the top screen assembly, remove the O-ring and place the screen in the pan bearing the corresponding mesh number.

i. Repeat steps g and h for successive filter screens.

j. When last separation is complete, dispose contaminated water according to approved safety procedures.

k. Dry screens and pans with diazodinitrophenol fractions in oven for a minimum of 12 hours.

l. Cool screens, pans, and diazodinitrophenol in desiccator for a minimum of 1 hour.

m. Weigh screens, pans, and diazodinitrophenol to nearest 0.001 g and record gross weight.

n. Subtract tare weight from gross weight to obtain weight of each diazodinitrophenol fraction.

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o. Plot weight-percentage versus screen size on graph paper. Plot 50 percent line of weight percentage and locate corresponding mesh size. This mesh size is the average particle size and shall meet the requirements of 3.4.

NOTE

Dispose of all residue and contaminated material according to approved safety procedures.

4.7.5 Explosive efficiency. An explosive efficiency test shall be made on dried and screened diazodinitrophenol from each lot by charging 200 ± 5 mg of diazodinitrophenol into five bronze Instadet shells (H-14-02) containing plant charged (0.23 g each) and pressed (0.22 ± 0.01 inch) pentaerythrite tetranitrate base charges. The plug shall be pressed to a height of 1.000 ± 0.003 inch. The caps are fired on 0.125 ± 0.005-inch-thick lead plates and the resultant holes in the plates are measured with a tapered pin gage. The requirements of 3.5 shall be met.

4.7.6 Bulk density. A bulk density measurement shall be made on dried and screened diazodinitrophenol from each lot. A shell with total volume of 3.298 cubic centimeters (cm^3) (0.475" ID x 1.135" DP) is filled to overflowing with loose diazodinitrophenol. The excess is scraped away, flush with the shell. The diazodinitrophenol in the shell is weighed and reported as the bulk density. The test is repeated to yield two measurements for each sample. The requirement of 3.6 shall be met.

4.7.7 Lot rejection. Unless otherwise specified in the contract or purchase order (see 6.2), if the composite sample representative of the lot fails to pass the inspection tests, the entire lot shall be rejected.

4.7.8 Workmanship. The product shall be visually inspected for the presence of foreign material during sampling and during the microscopic examination (see 4.7.3).

4.7.9 Inspection of packaging. The packaging, packing, and marking shall be examined for conformance to section 5. Critical characteristics shall undergo 100 percent inspection. Major and minor characteristics shall be examined using AQLs of 0.25 percent and 1.00 percent, respectively.

5. PACKAGING

5.1 Packing. The diazodinitrophenol shall be packed in accordance with 49 CFR 173.70 (October 1, 1989 revision). All bags shall conform to MIL-B-82647. The following requirements shall be met:

a. The diazodinitrophenol shall be packed wet with not less than 40 percent water by weight (C1) (see 6.7).

b. Bags shall be securely closed (M107).

c. Bags shall be uniformly constructed and free of pinholes, tears, cuts, sharp creases, wrinkles, or other imperfections which might impair usefulness (M108).

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5.2 Packaging. The diazodinitrophenol shall be packaged in accordance with 49 CFR 173.70 (October 1, 1989 revision). The following requirements shall be met:

a. The steel barrel or drum shall not leak or show signs of leakage (C2).
b. The steel barrel or drum shall be properly closed and sealed (M109).
c. Exterior of metal containers shall be free of bare spots other than slight scratches (201).

5.3 Marking. Marking shall be in accordance with 49 CFR 173.70 (October 1, 1989 revision). The following requirements shall be met:

a. Marking shall be identifiable and non-misleading (C3).
b. Marking shall be free of blurs (202).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The diazodinitrophenol is intended for use in loading fuze detonators and in the manufacture of primary compositions.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Amount of diazodinitrophenol to be procured.

d. Whether a first article inspection is required (see 3.1).

e. The facility designated to perform first article inspection (see 4.3).

f. The facility designated to perform quality conformance inspection, if other than the contractor (see 4.1 and 4.4).

g. Whether the provisions of MIL-A-48078 apply (see 4.4.1).

h. Quality conformance inspection lot and sample size (see 4.4.1 and 4.4.3).

i. Sampling plan and procedures (see 4.4.4).

j. Inspection equipment, calibration, design, and approval (see 4.6.3 and MIL-STD-45662).

k. Safety precautions (see 4.6 and 6.4).

l. Whether recertification is required (see 6.6).

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6.3 Equivalent test methods. The test methods specified herein are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (in accuracy and precision) to the method given herein. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer. This description should include the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.4 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this document is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.5 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Recertification. When specified in the contract or purchase order (see 6.2), the lot of diazodinitrophenol should be recertified every 3 years against the examinations and tests listed in 4.7.

6.7 Freezing conditions. A mixture of denatured alcohol and water in a ratio adequate to prevent freezing during shipment and storage should be used.

6.8 Subject term (key word) listing.

Detonator
Explosive, initiating
Primer

Custodians:
Army - AR
Navy - OS

Preparing activity:
Navy - OS
(Project 1376-0464)

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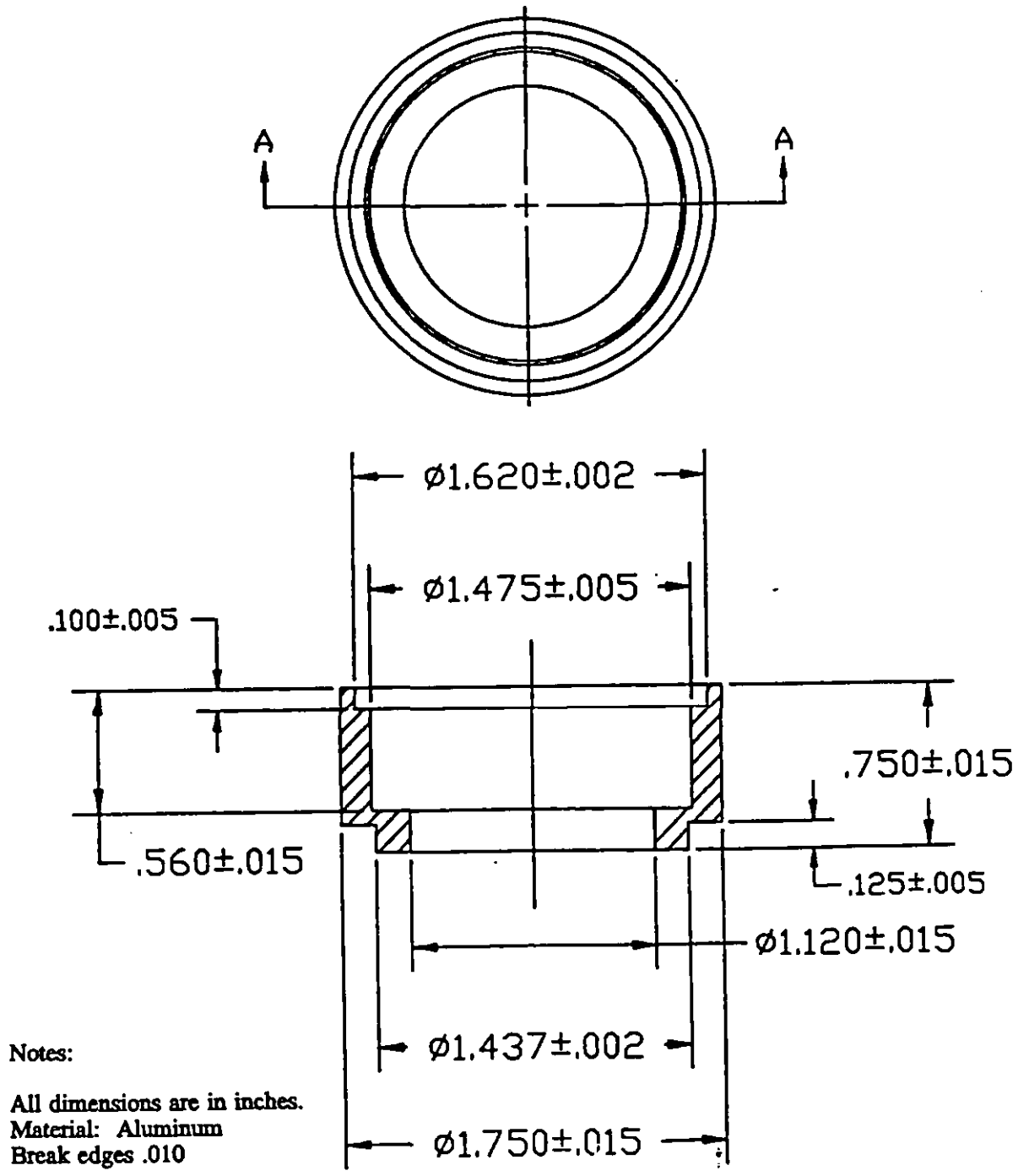
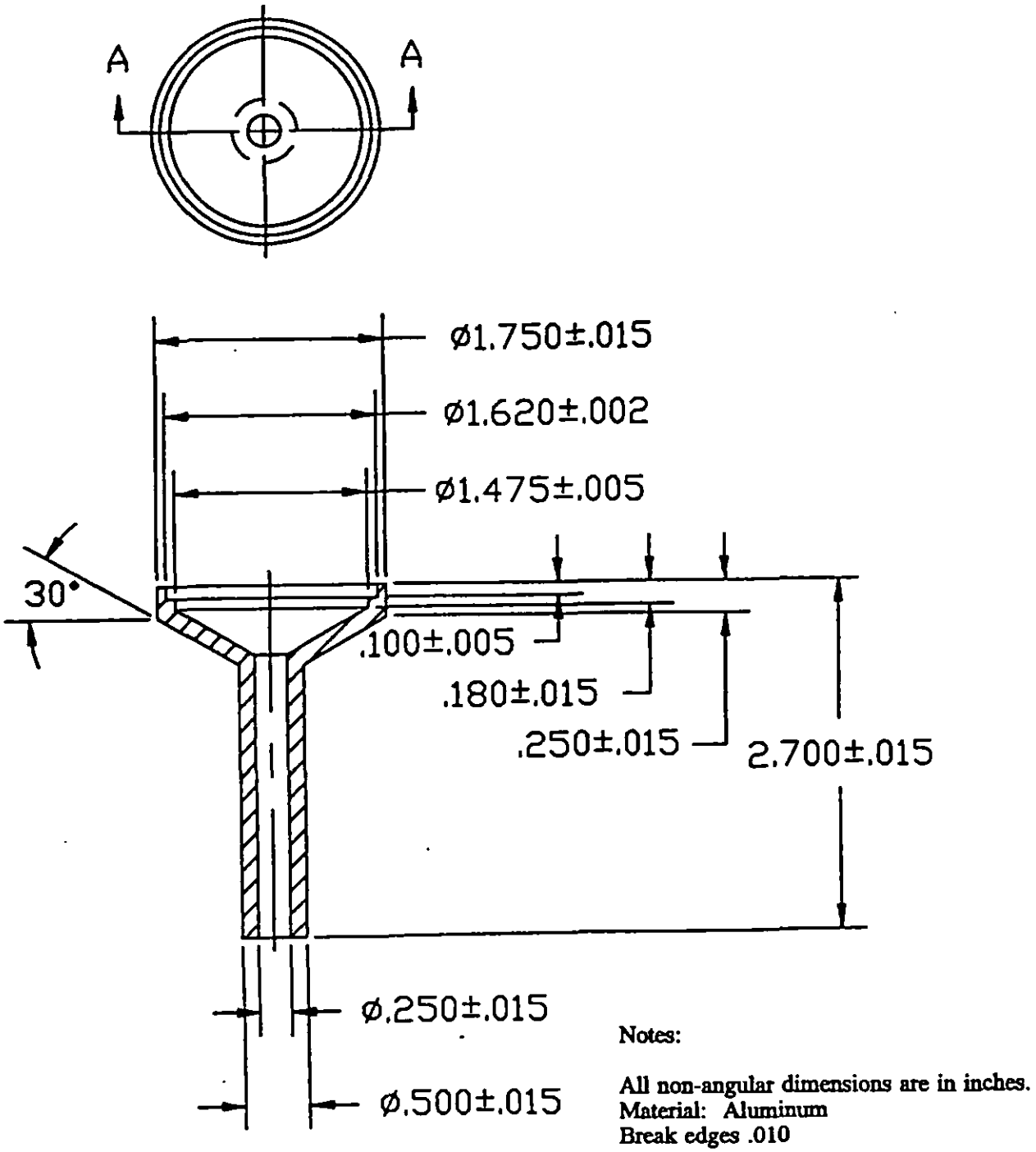


FIGURE 1. Screen frame.

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Notes:

All non-angular dimensions are in inches.
Material: Aluminum
Break edges .010

Section A-A

FIGURE 2. Screen funnel.

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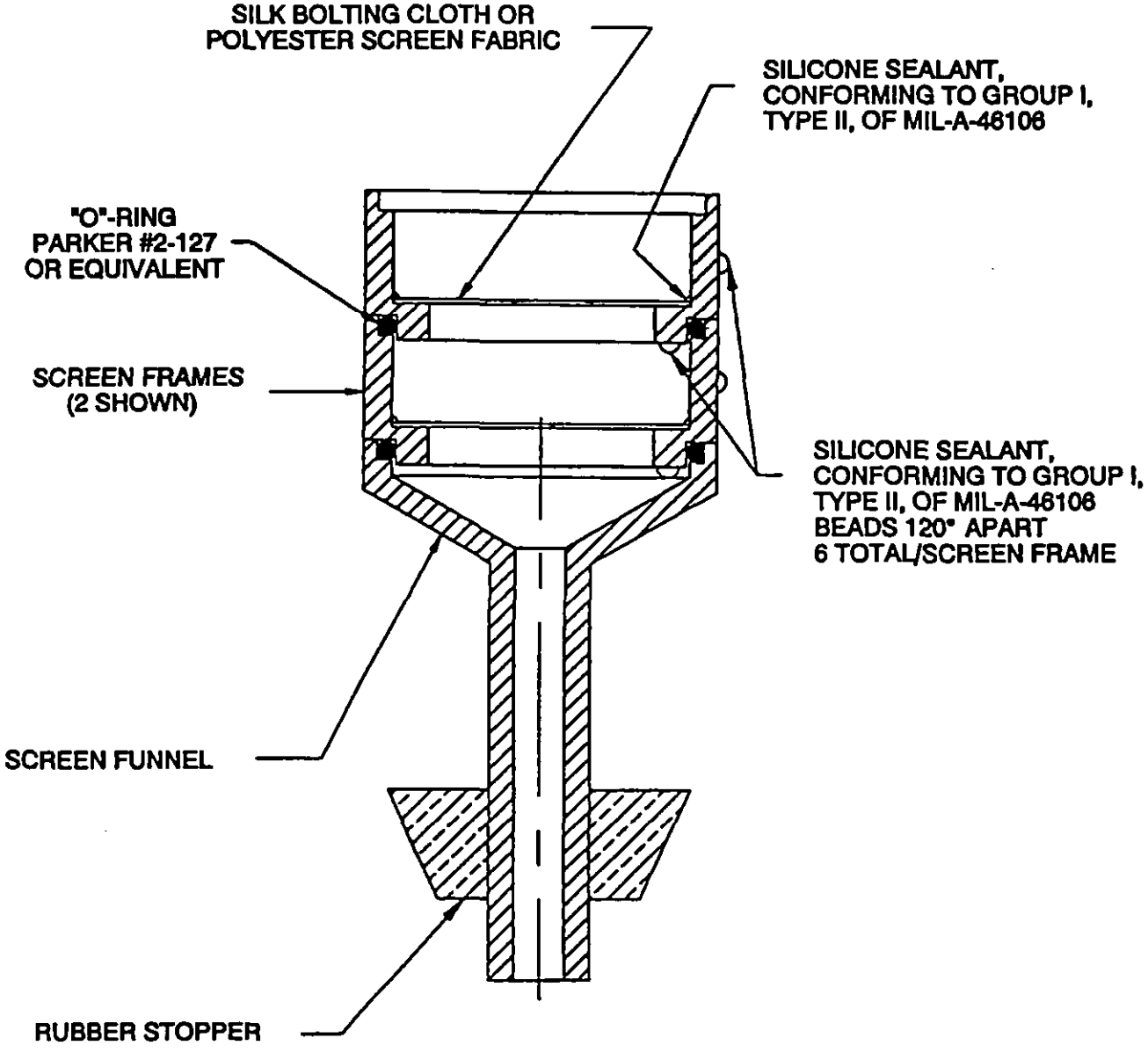


FIGURE 3. Screen assembly.

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-82885	931027

3. DOCUMENT TITLE

DIAZODINITROPHENOL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	(1) Commercial 301-743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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